

THE MAZE RUNNER

Targets:

- Integrate all concepts from Stage 2 into a functional robot.
- Select and integrate appropriate sensors, actuators, and control components to ensure reliable robot performance.
- Test, debug, and ensure proper functionality of the robot.

Key Components:

- Microcontroller.
- 2* DC motors.
- Motor driver.
- 3* any distance sensor (front, right and left).
- IMU
- Batteries.

***Note:** The listed components are the main components for the project, but you may use alternatives if available — any extra features you need to add (it's recommended) stick to it.*

Project Description & Requirements:

In a world where only the fastest and most precise survive, your challenge is to design a robot capable of navigating and solving a maze with maximum speed and efficiency. Your robot must detect obstacles, plan its path, and adapt to the maze autonomously while demonstrating reliable performance and accurate component integration.

Note that:

- The project will be on **software only**, you will not make it on the real life.
- The dimensions of the PCB must not exceed 11 cm * 11 cm.

Bonus:

- (+5 Pts) For the smallest robot.
- (+5 Pts) If you were to monitor the battery's percentage, how would you do that? include it in your design.
- (+2 Pts) Place 3D of your components to your PCB board on Altium.

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Submission:

- You must give a presentation to highlight your outstanding projects. You have to abide by these guidelines:
 - **Each** member in the team must participate in the presentation
 - **All calculations** circuit must be included and well explained.
 - **Videos and pictures** must be used to document your work.
- A PDF that contains:
 - All track width calculations.
 - Power calculations of all components.
 - Microcontroller selection justification (Which one did you choose and why?).
 - Battery lifetime (include all calculations concerning battery).

Resources:

[Examples on a micromouse robot](#)

[Micromouse international competition](#)

Evaluation:

- Robot functionality → 10 pts
- Calculations PDF → 10 pts
- Schematic layout (quality) → 5 pts
- PCB components placement and routing → 10 pts
- Presentation → 15 pts
- Teamwork → 10 pts

Total: 60 pts (+12 Bonus)

“The winner takes it all”



Deadline:
(8 DAYS)

